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Supplemental Information

High-Throughput Multiplexed Peptide-Centric Profiling Illustrates Both Substrate Cleavage Redundancy and Specificity in the MMP Family

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Supplemental Section

Supplemental Table S1, related to Figure 2. The raw cleavage data of 18,583 individual peptides in a Z-score format. Abbreviations in the "Construct" column: BF, substrates of the secretoty metalloproteinase (fragilysin) from *Bacteroides fragilis*; EP, control peptides that represent the common tag sequences of the DNA-peptide fusions; F and MAS, control peptides; MMB and MMS, control peptides that are selected from the CutDB, a proteolytic event database (http://cutdb.burnham.org) and MEROPS, the Peptidase Database, and that are expected to be cleaved by certain MMPs; MR, negative control peptides; PC, substrates of caspases; PD and PW, substrates of the NS2B-NS3 proteinase from Dengue and West Nile viruses, respectively; PE, substrates of enterokinase; PF, substrates of furin; PK, kinase substrates; PT, substrates of thrombin; S, control peptides; SFF, potential substrates of furin-like proteinases; MMPA, potential substrates of MMPs designed *in silico* using random number generator; SMMPT, potential MMP cleavage sequences derived from the crystal structures of human secretory proteins.

Supplemental Table S2, related to Figure 6. Comparison of the k_{cat}/K_M values of the peptide substrates tested by Turk et al., 2001 [Turk, B.E., Huang, L.L., Piro, E.T., and Cantley, L.C. (2001). Determination of protease cleavage site motifs using mixture-based oriented peptide libraries. Nat Biotechnol 19, 661-667] with the PWM scores. The original and normalized kcat/KM values measured by Turk et al., 2001 for the MMP-2, MMP-9 and MMP-14/MT1-MMP substrates are compared with the PWM scores calculated by the CleavPredict software.

Supplemental Fig. S1, related to Figures 1 and 2, and Table 1. Experimental assessment of the PWM predictions: the PMW score for the peptides substrates with the experimental Z-score over 2.5 for the individual MMPs. The horizontal lines are located the MMP dependent PWM threshold, above which the prediction score denotes the efficient substrates (blue dots). The red dots indicate the peptide sequences that are not recognized by the PWM scoring method.

Supplemental Table S2

MMP-2				
Peptide sequence	Original PWM score	Original k_{cat}/K_M	Normalized PWM score	Normalized k_{cat}/K_m
PQG-IA	3.91	10,100	0.49	0.12
PMS-MR	5.94	24,000	0.74	0.29
PVS-LR	7.98	82,000	1	1
PFS-MI	5.2	4,600	0.65	0.06
PLS-LT	6.39	13,200	0.8	0.16
PLS-LY	5.14	61,000	0.64	0.74
PES-LR	3.43	24,000	0.43	0.29
MMP-9				
PQG-IA	2.77	8,400	0.34	0.16
PMS-MR	7.02	51,000	0.87	1
PVS-LR	7.42	11,500	0.92	0.23
PFS-MI	6.32	21,000	0.79	0.41
PLS-LT	8.03	20,000	1	0.39
PLS-LY	7.83	49,000	0.98	0.96
PES-LR	6.53	12,600	0.81	0.25
MMP-14/MT1-MMP				
PQG-IA	2.21	3,600	0.25	0.35
PMS-MR	7.2	6,100	0.8	0.59
PVS-LR	7.21	4,300	0.8	0.42
PFS-MI	2.03	3,700	0.23	0.36
PLS-LT	6.4	10,300	0.71	1
PLS-LY	4.73	5,500	0.53	0.53
PES-LR	9	6,900	1	0.67

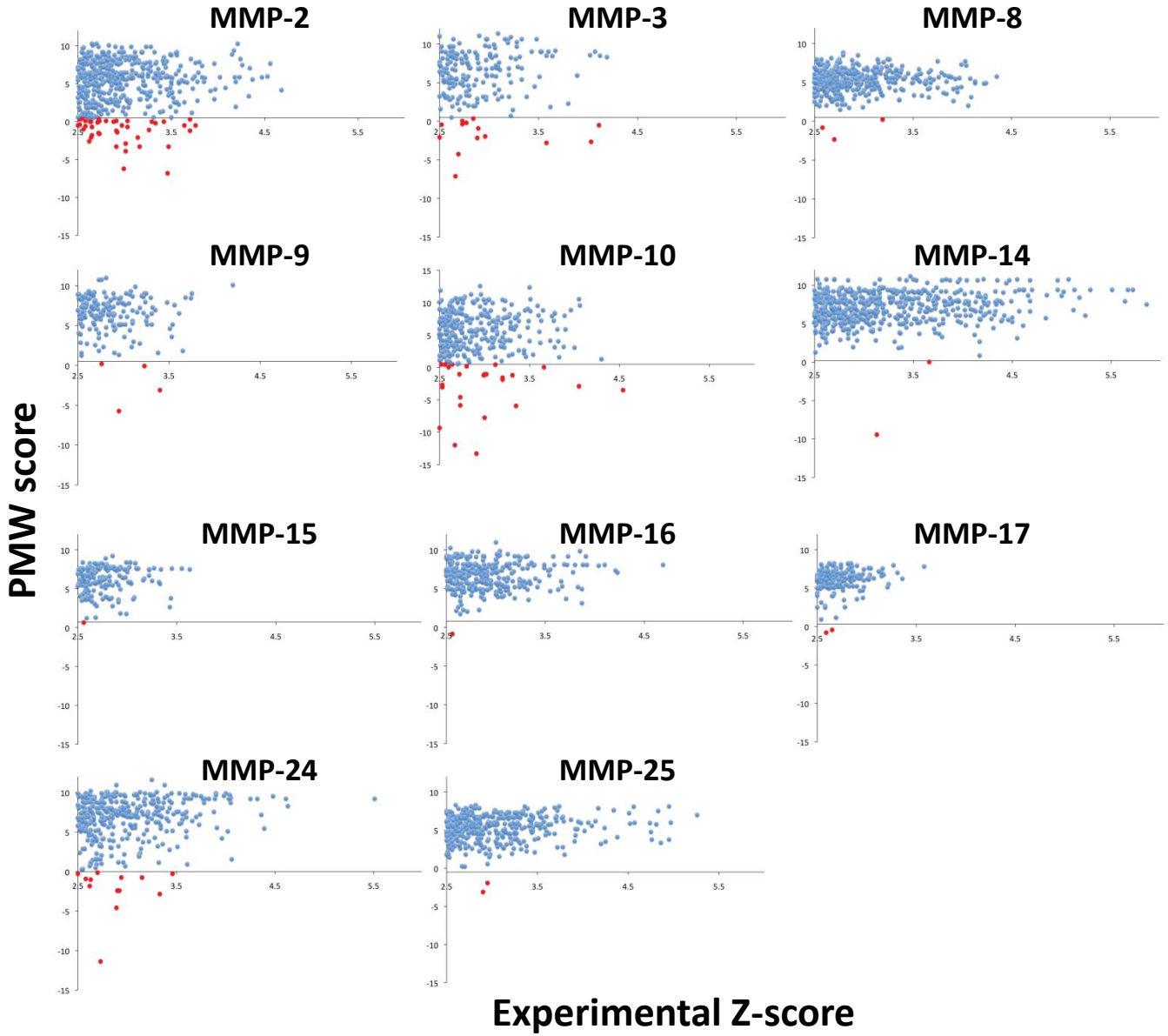
MMP-2
MMP-3
MMP-8
MMP-9
MMP-10
MMP-14
MMP-15
MMP-16
MMP-17
MMP-24
MMP-25
PMW score
Experimental Z-score

Fig. S1